CREDIT MITRA SCORING PLAN

SCORES are kept in the range 300-900 which follows the industrial standard of banks which they already understand like the CIBIL/FOIR scores etc.

The scores are made up of two major blocks :

- **Cash flow block**-this includes the FOIR analysis,salary regulatory,balance buffer,bounces,overdrafts these are basically the classical transaction based underwriting signals.
- **Lifestyle Block**- this is a six indices(L1-L6) derived from 24 basic merchant derived spendings,and each contributes the weight_i × (L_i − 50)/50 × max_points_i points.
- The scorecard method is used for the fact to keep the explanability of every output is auditable and the for the fact that it can be validated since **an opaque model can only be validated by its output, an additive scorecard can be validated factor-by-factor**

Two different kinds of "derived from"

Type 1 - the *measurement* is mathematically defined and research-validated. Things like cosine similarity for persistence, Shannon entropy for diversity, the burstiness statistic these are established mathematical constructs, borrowed directly from the cited papers' .

Type 2 - the *weights and thresholds* (0.6/0.4, ×5, ×10, the 0.35 income cutoff, archetype cutoffs like L1≥60) are set, *predictive validation (AUC/KS vs real defaults) via testing is future work to get labeled outcomes.*

The structure and the individual measurements are research-derived; the combination weights are priors, explicitly flagged as provisional pending real-outcome calibrationthis is the standard scorecard-development lifecycle.

Stage 0 - PDF upload → parsing (parsing.py)

parse_amount, parse_date, derive_direction turn raw statement text into structured Transaction rows (amount, date, credit/debit direction).

Stage 1 - Merchant enrichment (the DP-SLM already benchmarked)

Each debit narration (e.g. UPI/swiggy@axl/…) gets resolved via the fine-tuned payee-extraction model into a canonical merchant + category + is_essential + risk_flag + lifestyle_dim + recurring_type. **This is the upstream task for everything downstream from the computation of** L1–L6 from raw narration strings which require merchant-level resolution first.

Stage 2 - Feature engineering (build_features)

Computes :

- FOIR (Fixed Obligation to Income Ratio),
- salary regularity,
- essential/discretionary ratios,
- bounce/BNPL/gambling flags

this feeds the cash-flow block.

FOIR is a standard regulatory underwriting ratio used by RBI-regulated lenders in India (a debt-service-to-income cap banks already apply).

The general cashflow block is required as claimed in the Malaysia MSME paper (AUROC 0.821 vs 0.672, interpretable logistic beating black-box models) for the thin-file borrowers.

Stage 3 - Lifestyle profile (build_profile) the L1–L6 indices

For example we have a statement of 6 month transactions.

L1 - Essential Stability:

- essential_share = 0.55 (55% of debits went to kirana/electricity/rent/recharge)
- persistence = mean(cosine_similarity(c_m, c_{m+1})) across the 5 consecutive month-pairs = 0.72

L1=100 × (0.6 × essential_share + 0.4 × persistence)

L1 = 100 × (0.6 × 0.55 + 0.4 × 0.72) = 100 × (0.33 + 0.288) = 61.8

Basis for the *measurement*: Persistence is defined as the cosine similarity between consecutive months' category-spend vectors featured from the EPJ paper's temporal-dimension framework (Gladstone et al., dimension 2) in which they define persistence this way and show it correlates with self-control.
The constants 0.6,0.4 have been directly used from the paper however these numbers can be changed and rectified during testing.

L4 - Commitment Index:

recurring_merchants detected: Zerodha SIP (₹5,000/mo), LIC premium (₹2,000/mo), rent (₹12,000/mo)

Σ ₹recurring = 19,000/month; income = 45,000/month

raw_ratio = min(1, 19000/(0.35×45000)) = min(1, 19000/15750) = min(1, 1.206) = 1.0 (capped)

regularity_factor = 1 − normalized_std(payment day-of-month deviations) = 1 − 0.08 = 0.92

L4 = 100 × 1.0 × 0.92 = 92

Basis for the *measurement*: detecting same-merchant periodicity (≥3 near-monthly payments) as a proxy for voluntary sustained obligation is directed by the EPJ paper's finding that self-control shows up as regular, low-variance transaction patterns - L4 is your operationalization of their psychological construct into a rule you can compute from merchant-tagged transactions.
Basis for the *0.35 threshold*: this loosely tracks the idea that recurring voluntary commitments shouldn't consume more than roughly a third of income if the person is to be considered "in control" - it's calibrated against common Indian debt-service norms (RBI-linked FOIR guidance typically caps *total* obligations around 40–50%; 0.35 for voluntary recurring spend alone, before EMIs, is a conservative expert sub-allocation).

L2, L3, L5, L6 follow the same pattern - a ratio of merchant-tagged spend to total spend, with a measurement basis (share-of-spend is a standard descriptive statistic) and an set multiplier.

The behavioral-texture metrics deserve their own callout because two of them *are* rigorously defined, independent of any weight-tuning:

- **Category diversity** = normalized Shannon entropy of the category-spend distribution - a well-established information-theoretic measure (Shannon, 1948), used unmodified.
- **Burstiness** $B = (\sigma-\mu)/(\sigma+\mu)$ of inter-transaction time gaps - this is the Goh–Barabási burstiness parameter from human-dynamics/network-science literature (Goh & Barabási, *"Burstiness and memory in complex systems,"* EPL 81, 48002, 2008). It's bounded in [−1, 1]: B=−1 is perfectly regular (periodic), B=1 is maximally bursty (long silences punctuated by clusters).

Stage 4 - Archetype assignment

Rule-based, first-match-wins over the six index values (e.g. salary detected and L1≥60 and L4≥50 and L5≥70 → *Disciplined Urban Salaried*).

Basis: these threshold cutoffs are set but the **validation is made sure by the persona test** such that the six synthetic personas namely stable_salaried, gig_irregular, over_leveraged, frequent_bouncer, cash_heavy, disciplined_saver are constructed in such a way that they map to their intended archetypes and rank in the intuitively-correct order.

Stage 5 - Two-block scorecard combination

$$\text{score_contribution}_i = \text{weight}_i \times (L_i - 50)/50 \times \text{max_points}_i$$

This centers each index at 50 (neutral), so an index above 50 adds points and below 50 subtracts, scaled by that index's max point allocation.

Basis: this (value − midpoint)/range normalization before weighting is the standard construction pattern in WOE and additive scorecards which follow the distributed proportionality to how far an applicant deviates from a neutral baseline. Summed with the cash-flow block, banded into 300–900.

Stage 6 - Report output

Score + band + archetype + per-factor breakdown, tagged by block for user to understand the explanability of the outputs given out and with the *AUC, KS, PSI measures we can validate all the formulas we formulate in the pipeline while testing*

References

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4. *Cash Flow Underwriting with Bank Transaction Data* (Malaysia MSME), arXiv:2510.16066 (2025) - backs cash-flow-over-bureau-data + interpretable-beats-blackbox
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7. Shapley-value interpretable scorecards, PMC11318906 - backs factor-level explainability
8. Account Aggregator framework & financial inclusion in India - backs the data-access model
9. CFPB cash-flow data blog - regulatory framing precedent
10. LightGBM credit risk, arXiv:2308.08762 - flagged future ML path once labels exist